

HUY B NGUYEN

Santa Barbara, CA | (559) 905-2116 | huy_nguyen@ucsb.edu | [linkedin.com/in/huynghuy06](https://www.linkedin.com/in/huynghuy06) | github.com/huynghuy9999 | US Citizen

EDUCATION

University of California - Santa Barbara

Bachelor of Science in Electrical Engineering

Santa Barbara, CA

June 2028

- **Honors/Awards:** Amazon Future Engineer (1 of 400 nationally), Promise Scholar (UCSB's Full scholarship), Elks Scholar (1 of 480 nationally)
- **Coursework:** Signals & Systems, Analog Circuit Analysis, Linear Algebra, Vector Calculus, Probability & Stats, Classical & Quantum Mechanics
- **Clubs:** Collegiate Tennis, Gaucho Rocketry, Sigma Eta Pi, Ascend Consulting Group, Association for Computing Machinery

EXPERIENCE

Software Development Engineer Intern

Amazon (Alexa)

June 2026 - Present

Seattle, WA

- Designed a real-time AI character calling system from scratch via WebRTC and brought to life a feature actively requested by 15,000+ Alexa customers
- Built a custom audio engine resolving a 4.7x speed mismatch between AI speech generation and hardware playback to achieve sub-1.5s conversational latency
- Fixed a bug in the platform's SDK then extended the model to support conversation memory, interruption handling, and failure recovery — none existed natively
- Shipped live audio, a 2D animated avatar with lip-sync, and automated content safety on every sentence; fully integrated to Amazon's pre-production infrastructure

Product & Mobile App Software Engineer

Polymarket

April 2026 - June 2026

Remote

- Partnered with Polymarket's Head of Product, a Senior Researcher, and Engineering team to translate 300+ user interviews into mobile feature proposals
- Identified onboarding friction of Polymarket's 45% non-trader segment to shape feature directions for the world's largest prediction market
- Designed & pitched concepts like a trading Heat Map & an "Analyst vs. Casual" mobile interface potentially serving hundreds of thousands of users monthly

Software Development Engineer Intern

Amazon (Alexa)

June 2025 - September 2025

Seattle, WA

- Reduced GenAI hallucination by fine-tuning internal LLMs on proprietary datasets, improving response consistency across multiple types of user queries
- Built a customer-facing Alexa+ privacy feature by integrating 3 backend services into a unified API for natural-language access to call history data via voice requests
- Designed an LLM evaluation framework to validate non-deterministic model behavior across 50+ test scenarios & improve pre-deployment reliability
- Shipped a working POC via AWS Gamma on a system designed to support ~50M requests/week to validate E2E performance and scalability

PROJECTS

iGauchoBack - UC Degree Roadmap | Next.js, TypeScript, Supabase, React Flow, Upstash, Sentry

<https://hackathon-nu-taupe.vercel.app/>

May 2026 - Present

- Shipped a UC degree-planning platform with 72+ interactive roadmaps, helping students visualize degree paths without relying on manual advising
- Engineered a TypeScript ingestion pipeline that converts inconsistent UC course catalogs into validated prereq graphs across multiple institutions
- Hardened backend APIs with rate limiting, edge caching, and Sentry observability to keep the platform responsive and reliable under concurrent use

Orbit AFE | JavaScript, CSS, SQL, Vercel, Supabase, Leaflet, Globe.gl

<https://orbitafe.vercel.app/>

July 2026 - July 2026

- Launched Orbit in 24 hours, connecting interns across 25+ North American hubs through a live 2D/3D map with real-time presence, opt-in location privacy
- Architected real-time DMs, group chat, and nine-signal matching that converts profile data into ranked introductions to keep discovery-to-connection entirely in-app

Chromatic Guitar Tuner | Python, SoundDevice, NumPy

April 2026 - May 2026

- Built a real-time low latency (~46ms), high-precision audio tuner that enables pitch detection across all 12 musical notes
- Achieved sub-Hz tuning precision by implementing signal-processing techniques including Harmonic Product Spectrum, FFT, etc
- Engineered a thread-safe audio pipeline to stream live microphone input without blocking real-time pitch analysis

LiDAR Home Security Detection System | Arduino Uno, C++, I2C

September 2024 - December 2024

- Built a servo-driven TOF10120 LiDAR security system that scans 180° at every 1° to store baseline distance and pinpoint unexpected movement up to 170cm, recognized at Northrop Grumman project expo
- Hardened false-alarm prevention by combining a 100cm proximity filter with angle-based distance comparisons and sensor retry logic before triggering detection
- Delivered precise directional alerts that identify the exact degree of detected movement via angle-indexed array lookup during continuous LiDAR sweeps

LEADERSHIP

Vice President of Finance

Sigma Eta Pi - Professional Fraternity, UCSB

Oct 2025 - Present

- Scaling membership from inception to 60+ active members in two quarters (200% growth) & established a structured 7-week founder education program
- Owning organizational finances end-to-end, including sponsorship acquisition, budgeting, and reporting compliance with national leadership standards

1 of 2 Design & Make Ambassador @ UCSB

Autodesk

February 2026 - Present

- Building the program's campus presence for the 1st time from scratch, reaching 100+ students & securing 4 talks across engineering + business/design orgs

SKILLS & INTERESTS

Languages: Java, C++, Python, Typescript, MATLAB, Groovy, Javascript, HTML/CSS

Embedded & Hardware: Arduino Uno, I2C, LTSpice, Oscilloscope, Soldering, Digital Multimeter, Fusion 360, Onshape, Revit

Tools/Frameworks: WebRTC, REST, GStreamer, EC2, S3, SSM, ElevenLabs API, Spring, Supabase, Mockito, Microservices, Pandas, NumPy, Maven/Gradle, CI/CD, Git

Hobbies/Interests: Drag Racing, Rock Crawling, Sports Photography, Reading, Sticker Collecting, Market Analysis, Speed Cubing